Transient Low-Order Cues Bootstrap Higher-Order Compositional Representations in Transformers

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Abstract

We study whether training-distribution geometry can alter formation of a higher-order predictive representation while the target conditional, irreducible noise, marginals, and total predictable information remain fixed. In a finite-field parent-exposed task, two uniform parents determine a modular center and a fixed additive-noise channel determines the target. Version 3.5 provided exploratory evidence that transient full-support low-order cues advanced predictive-state formation. We prospectively froze the architecture, optimizer configuration, objective, evaluation protocol, onset definition, and statistical test before confirmatory training. The v3.6 study was a prospectively frozen confirmatory replication using held-out rules, cue permutations, and initializations, with three cue doses. Verified onset occurred in 1/8, 6/8, and 8/8 blocks. The prospectively frozen one-sided exact blocked Page-style test enumerated 1,679,616 assignments and gave $p = 9.5259869e-06$. All 13 nonzero-cue cells positive at cue removal retained the representation threshold at 6M and 10M. Temporary cue-bearing distributions therefore causally changed onset timing or probability within the task and model family under the prospectively frozen protocol. The experiment does not identify singleton mutual information alone as causal and does not establish a universal Transformer property.

1 Introduction

The central question is: *How does training-distribution geometry affect the accessibility of task-sufficient representations when the final task information is fixed?* A representation may be statistically available yet difficult to discover because useful optimization directions are weak, cancel across examples, or become accessible only after other features organize. We therefore intervene on early training-distribution geometry while holding the target conditional and information budget fixed.

Project Kepler uses a finite-field mechanism assay motivated by, but distinct from, its recurrent production task. The parent-exposed design intentionally separates (1) identifying relevant variables, which the interface controls by presenting both parents directly, from (2) composing discovered variables into a predictive representation, which remains the learning problem. The present paper does not claim that the recurrent task has been solved.

Version 3.5 is exploratory discovery. Before confirmatory training, all outcome definitions and analysis choices were prospectively frozen. Version 3.6 is a prospectively frozen confirmatory replication using held-out rules, cue permutations, and initializations. The discovery data are excluded from the primary confirmatory p-value. The aim is neither a new scaling law nor a universal intelligence principle; it is a bounded causal test within one controlled task and model family. Figure 1 summarizes the construction and cue schedule.

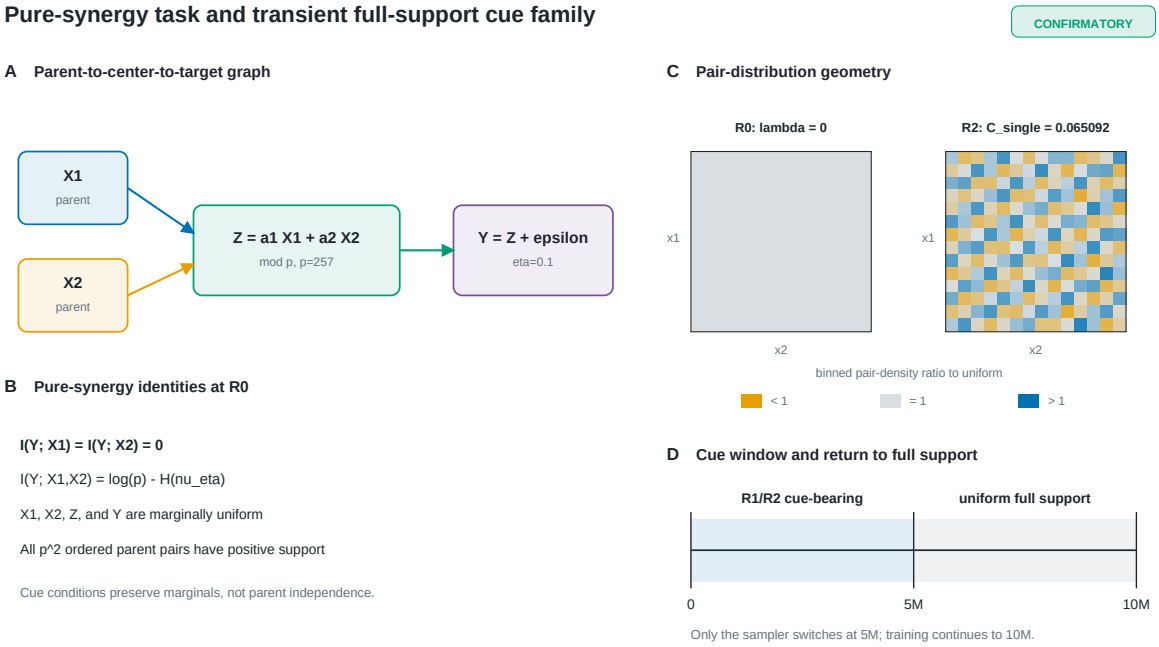


Figure 1: Controlled task, pure-synergy identities, R0-versus-R2 pair-distribution geometry, and transient cue schedule. Binned pair-density ratios and all displayed values are generated from frozen distribution JSON.

2 Related Work

Grokking and delayed generalization. Modular-arithmetic grokking studies established that small neural networks can generalize long after achieving near-perfect training performance under a fixed training distribution Power et al. (2022). Follow-up work connected delayed generalization to structured representation learning and its training dynamics Liu et al. (2022). This paper does not offer a universal explanation of grokking. It instead asks whether the geometry of the *early* training distribution, with the final conditional and information budget held fixed, changes the accessibility or timing of a delayed representational transition.

Mechanistic interpretability of modular arithmetic. Mechanistic analyses of modular addition have recovered Fourier-like internal representations and circuits implementing trigonometric structure Nanda et al. (2023), while other architectures and initializations can learn qualitatively different modular-addition algorithms Zhong et al. (2023). Here, fixed-Fourier probes are diagnostics alongside linear and fixed-capacity MLP probes. Probe accuracy establishes extractability under the specified probe, not causal use by the model Hewitt and Liang (2019); Belinkov (2022). We do not assume that a specific Fourier circuit must be learned, and the causal result concerns the timing or probability of task-sufficient representation formation rather than identification of a unique circuit.

Curriculum learning. Classic curricula order examples by difficulty Bengio et al. (2009); Hacohen and Weinshall (2019); self-paced methods adapt selection to the learner Jiang et al. (2015);

and continuation methods smooth or deform an optimization problem Mobahi (2016). The present intervention is a *distributional curriculum* whose distinctive control is preservation of the target conditional, Bayes entropy, total predictable information, and full pair support. It changes the geometry of a full-support sampling distribution during an early window. We call this narrower intervention *distributional bootstrapping of representation formation*.

Optimization path dependence. Learning dynamics can show long plateaus, rapid transitions, and strong dependence on initialization even in analytically tractable deep linear networks Saxe et al. (2014). This motivates studying training paths, but it does not by itself identify the mechanism in the present Transformer. Our gradient statistics are therefore treated as descriptive correlates rather than causal mediators.

Partial Information Decomposition. Partial Information Decomposition separates redundant, unique, and synergistic contributions in multivariate information Williams and Beer (2010). Multiple operationally motivated decompositions exist Bertschinger et al. (2014), so synergy language depends on the adopted definition. The full-support base task is a pure-synergy construction only in the limited operational sense needed here: singleton predictive information is zero while joint predictive information is nonzero. We do not attempt a full PID decomposition of the cue distributions or learned representations.

3 Controlled finite-field task

The field size is $p = 257$. The predictive center is $z = a_1x_1 + a_2x_2 \bmod p$, and $y = z + \epsilon \bmod p$ with $P(\epsilon = 0) = 1 - \eta$, $P(\epsilon = r) = \eta/(p-1)$ for $r \neq 0$, and $\eta = 0.1$. Thus $H(Y \mid X_1, X_2) = 0.879601$ nats, and the target conditional is unchanged across conditions. The primary metric is

$$R = \frac{\mathbb{E}_h D_{\mathrm{KL}}(q_h \| p_\theta)}{\log p - H(\nu_\eta)}.$$

A uniform predictor has $R = 1$ and the oracle has $R = 0$. Objective, evaluation, bootstrap, and probe details appear in Supplementary Methods.

4 Pure-synergy information structure

Under uniform full support, X_1, X_2, Z, Y are uniform, $I(Y; X_1) = I(Y; X_2) = 0$, and $I(Y; X_1, X_2) = 4.66948$ nats. The scalar cue index orders selected constructed distributions; it is not a unique-information percentage.

5 Full-support cue-family construction

For $z = a_1x_1 + a_2x_2 \bmod p$, the prospectively frozen cue family was

$$P_{\lambda,m}(x_1, x_2) = \frac{1-\lambda}{p^2} + \frac{\lambda}{mp}\mathbf{1}[(z - cx_1) \bmod p \in B_m],$$

where B_m is a frozen size-m subset induced by the cue permutation and c is its frozen cue coefficient. Cue dose was summarized by

$$C_{\mathrm{single}} = \frac{I(Y; X_1) + I(Y; X_2)}{I(Y; X_1, X_2)}.$$

Table 1: Every registered cue distribution retained strict full support and the frozen information identities.

Version	Status	Rule	Permutation	Condition	m	Target C_single	Achieved C_single	Lambda	Min pair probability	Max uniformity error	H(Y|X1,X2), nats	Joint-I abs. error	Strict support
v3.5	exploratory	45,36	v3.5	C0	64	0	7.608e-16	0	1.514e-05	9.107e-18	0.879601	1.066e-14	yes
v3.5	exploratory	45,36	v3.5	C1	64	0.016	0.016	0.182955	1.237e-05	2.602e-18	0.879601	0	yes
v3.5	exploratory	45,36	v3.5	C2	64	0.065092	0.065092	0.38055	9.379e-06	1.735e-18	0.879601	1.776e-15	yes
v3.5	exploratory	45,36	v3.5	C3	64	0.13	0.13	0.546166	6.871e-06	9.107e-18	0.879601	5.329e-15	yes
v3.5	exploratory	45,36	v3.5	geometry_concentrated	1	0.065092	0.065092	0.078826	1.395e-05	1.908e-17	0.879601	1.599e-14	yes
v3.5	exploratory	45,36	v3.5	geometry_diffuse	128	0.065092	0.065092	0.594889	6.133e-06	5.638e-18	0.879601	1.776e-15	yes
v3.6	confirmatory	205,5	permutation_0	R0	64	0	7.608e-16	0	1.514e-05	9.107e-18	0.879601	1.066e-14	yes
v3.6	confirmatory	205,5	permutation_0	R1	64	0.016	0.016	0.182955	1.237e-05	1.735e-18	0.879601	0	yes
v3.6	confirmatory	205,5	permutation_0	R2	64	0.065092	0.065092	0.38055	9.379e-06	1.301e-18	0.879601	0	yes
v3.6	confirmatory	205,5	permutation_1	R0	64	0	7.608e-16	0	1.514e-05	9.107e-18	0.879601	1.066e-14	yes
v3.6	confirmatory	205,5	permutation_1	R1	64	0.016	0.016	0.182955	1.237e-05	1.735e-18	0.879601	0	yes
v3.6	confirmatory	205,5	permutation_1	R2	64	0.065092	0.065092	0.38055	9.379e-06	1.301e-18	0.879601	0	yes
v3.6	confirmatory	123,209	permutation_0	R0	64	0	7.608e-16	0	1.514e-05	9.107e-18	0.879601	1.066e-14	yes
v3.6	confirmatory	123,209	permutation_0	R1	64	0.016	0.016	0.182955	1.237e-05	1.735e-18	0.879601	0	yes
v3.6	confirmatory	123,209	permutation_0	R2	64	0.065092	0.065092	0.38055	9.379e-06	1.301e-18	0.879601	0	yes
v3.6	confirmatory	123,209	permutation_1	R0	64	0	7.608e-16	0	1.514e-05	9.107e-18	0.879601	1.066e-14	yes
v3.6	confirmatory	123,209	permutation_1	R1	64	0.016	0.016	0.182955	1.237e-05	2.602e-18	0.879601	0	yes
v3.6	confirmatory	123,209	permutation_1	R2	64	0.065092	0.065092	0.38055	9.379e-06	1.301e-18	0.879601	0	yes

Table 2: Exploratory discovery outcomes decoded with the frozen onset and establishment rules.

Version	Status	Condition	n	Onset by 5M	Onset by 10M	Onset exposures (M)	Terminal ordinals	Established by 5M	Established by 10M	Mean R, 5M pre	Mean R, 10M
v3.5	exploratory	C0	5	0	0	—, —, —, —, —	8, 8, 8, 8, 8	0	0	0.973341	0.628411
v3.5	exploratory	C1	5	2	5	7.5, 5, 6, 6, 5	6, 4, 5, 5, 4	0	5	0.428643	0.025721
v3.5	exploratory	C2	5	5	5	4, 3, 3, 4, 3	3, 2, 2, 3, 2	5	5	0.013564	0.003252
v3.5	exploratory	C3	5	5	5	4, 3, 3, 4, 4	3, 2, 2, 3, 3	5	5	0.007108	0.00162
v3.5	exploratory	geometry_concentrated	5	5	5	4, 5, 4, 5, 4	3, 4, 3, 4, 3	5	5	0.014501	0.001528
v3.5	exploratory	geometry_diffuse	5	5	5	5, 3, 3, 3, 4	4, 2, 2, 2, 3	4	5	0.055923	0.007754

At $\lambda = 0$, the distribution is uniform full support. For every registered nonzero dose, all pair probabilities remain positive and both parent marginals remain uniform, although X_1 and X_2 need not remain jointly independent. Z and Y remain uniform; the target conditional, Bayes entropy, and joint predictable information remain fixed. C_{single} is an operational cue index, not a PID unique-information fraction, and it does not isolate singleton mutual information from other distributional geometry. Figure 1 compares R0 with one cue-bearing distribution. Table 1 reports the frozen audits; the derivation and deterministic root-solving protocol appear in Supplementary Methods.

6 Exploratory discovery experiment

Version 3.5 used rule $(45, 36)$, five paired seeds, four ordered $m = 64$ doses, and two matched-index geometry controls. Verified onset occurred in 0/5 C0, 5/5 C1, 5/5 C2, and 5/5 C3 seeds. The exploratory exact test gave $p = 1.6075103e - 05$. These observations selected frozen confirmatory conditions and definitions; they are not confirmatory data.

Figure 2 and Table 2 report the discovery trajectories and frozen outcomes.

7 Prospectively frozen confirmatory replication

Version 3.6 was a prospectively frozen confirmatory replication using held-out rules, cue permutations, and initializations. Crossing two held-out rules, two held-out cue permutations, and two held-out initialization seeds yielded eight paired blocks and 24 trajectories. R0, R1, and R2 shared initial weights and all non-condition random fields within each block. The primary inference unit was the eight rule × permutation × initialization blocks. The 24 trajectories were paired condition observations—three per block—not 24 independent inferential units. Verified onset occurred in 1/8, 6/8, and 8/8 blocks. The observed Page-style statistic was 110.0. Under the prospectively frozen sharp null, condition labels were exchangeable within each block. Exactly 16 of 1,679,616 assignments were at least as large; the exact upper-tail count included the observed assignment. This gave one-sided $p = 9.5259869e - 06$.

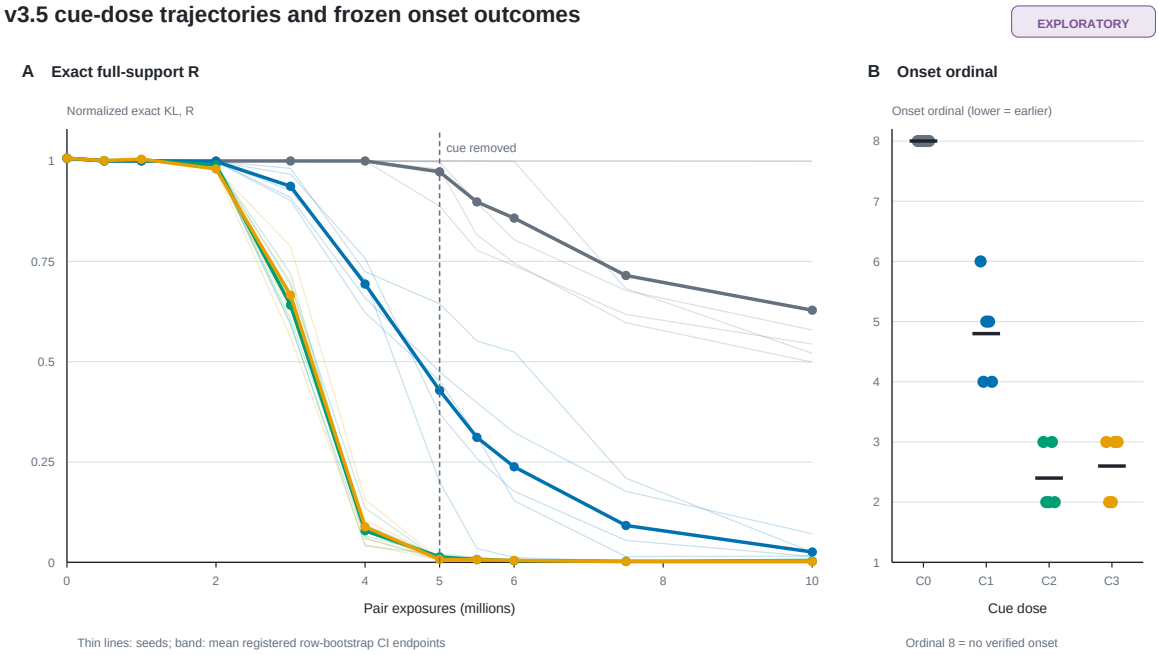


Figure 2: Exploratory v3.5 R trajectories and onset outcomes. Bands are means of registered row-bootstrap endpoints, not new group-level intervals.

Table 3: Prospectively frozen outcomes across eight paired blocks. Each block crosses one held-out rule, cue permutation, and initialization; trajectories are paired observations, not independent inference units.

Version	Status	Condition	n	Onset by 5M	Onset by 10M	Onset exposures (M)	Terminal ordinals	Established by 5M	Established by 10M	Mean R, 5M pre	Mean R, 10M
v3.6	confirmatory	R0	8	0	1	—, —, 7.5, —, —, —, —, —	8, 8, 6, 8, 8, 8, 8, 8	0	2	1.000239	0.432743
v3.6	confirmatory	R1	8	5	6	4, 4, 5, 4, 7.5, —, —, 4	3, 3, 4, 3, 6, 8, 8, 3	4	6	0.303076	0.110766
v3.6	confirmatory	R2	8	8	8	3, 3, 4, 4, 4, 3, 4, 4	2, 2, 3, 3, 3, 2, 3, 3	6	8	0.04371	0.005181

Figures 3 and 4 show the paired onset ordinals and exact null. Tables 3, 4, and 5 report confirmatory outcomes, exact tests, and paired differences. The transition-coding rule is specified in Supplementary Methods. Rule- and permutation-stratified tests are secondary, nominal, and not multiplicity-adjusted.

8 Representation persistence after cue removal

At 5M only the sampler changed for nonzero-cue runs. Zero-step checks verified unchanged outputs and model, optimizer, scheduler, exposure, and RNG state. All 13 threshold-positive R1/R2 cells remained positive at 6M and 10M, supporting learning-path preconditioning rather than a permanently required shortcut.

Figure 5 and Table 6 report the persistence result and zero-step switch audit.

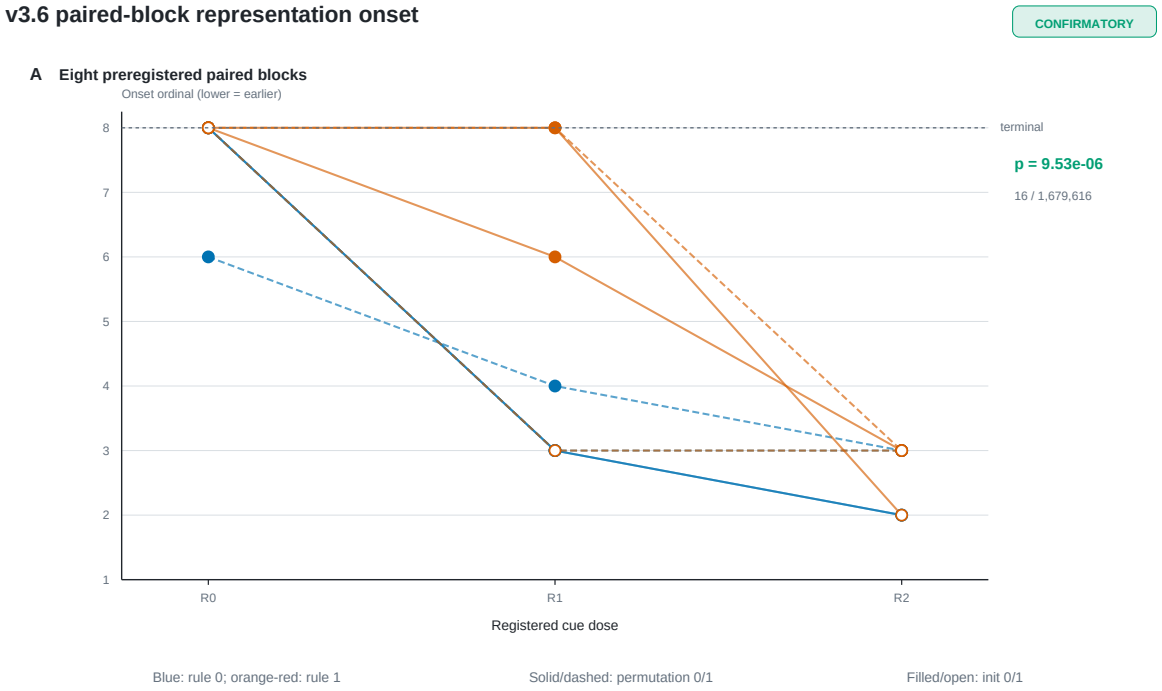


Figure 3: Paired terminal-coded onset ordinals for all eight confirmatory blocks.

Table 4: The primary row is confirmatory. Rule- and permutation-stratified rows are secondary, nominal, and not multiplicity-adjusted; v3.5 is excluded throughout.

Analysis	Inference status	Blocks	Exact assignments	Observed statistic	Upper-tail count	One-sided p	p<0.05	v3.5 included
primary	primary	8	1679616	110	16	9.526e-06	yes	no
rule_0	secondary_nominal_not_multiplicity_adjusted	4	1296	55.5	2	0.001543	yes	no
rule_1	secondary_nominal_not_multiplicity_adjusted	4	1296	54.5	8	0.006173	yes	no
permutation_0	secondary_nominal_not_multiplicity_adjusted	4	1296	55.5	2	0.001543	yes	no
permutation_1	secondary_nominal_not_multiplicity_adjusted	4	1296	54.5	8	0.006173	yes	no

9 Rule and cue-geometry heterogeneity

R1 onset occurred in 4/4 rule-0 and 2/4 rule-1 blocks, while R2 onset occurred in 4/4 and 4/4. The ordered effect spans both rules but the mid-dose response is rule-dependent. In exploratory matched-index controls, concentrated and diffuse geometries differed from $m = 64$, so equal cue-index values are not interchangeable.

Figures 6 and 7 separate the confirmatory rule comparison from the exploratory geometry comparison.

Table 5: Differences are higher registered dose minus lower registered dose within each v3.6 paired block.

Comparison	Blocks	Mean delta R, 5M	Median delta R, 5M	Mean delta R, 10M	Median delta R, 10M	Mean onset-ordinal delta	Median onset-ordinal delta
R1_minus_R0	8	-0.697162	-0.789907	-0.321977	-0.485173	-3	-3.5
R2_minus_R0	8	-0.956528	-0.98357	-0.427562	-0.528758	-5.125	-5
R2_minus_R1	8	-0.259366	-0.151302	-0.105585	-0.001297	-2.125	-1

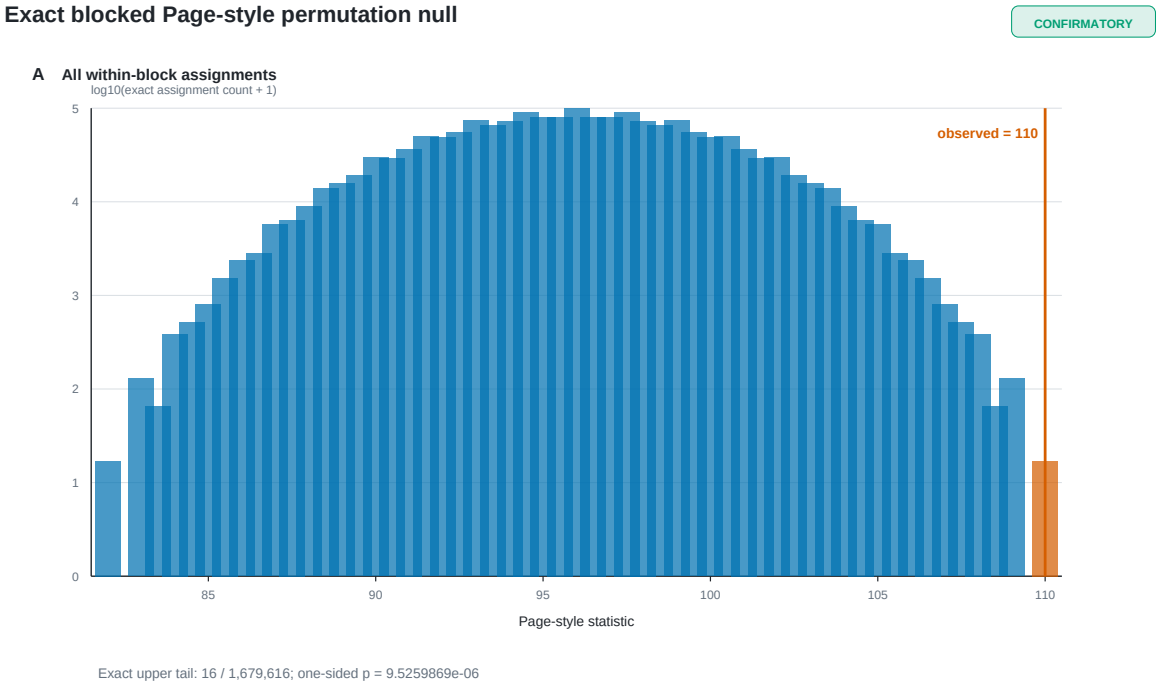


Figure 4: Exact v3.6 permutation null. Version 3.5 is excluded.

Table 6: Retention is evaluated only among cells positive at the 5M pre-switch checkpoint.

Version	Status	Condition	Positive at 5M pre	Retained at 6M	Retained at 10M	Fraction at 6M	Fraction at 10M	Sampler after 5M
v3.5	exploratory	C0	0	—	—	—	—	uniform_full
v3.5	exploratory	C1	2	2	2	1	1	uniform_full
v3.5	exploratory	C2	5	5	5	1	1	uniform_full
v3.5	exploratory	C3	5	5	5	1	1	uniform_full
v3.5	exploratory	geometry_concentrated	5	5	5	1	1	uniform_full
v3.5	exploratory	geometry_diffuse	5	5	5	1	1	uniform_full
v3.6	confirmatory	R0	0	—	—	—	—	uniform_full
v3.6	confirmatory	R1	5	5	5	1	1	uniform_full
v3.6	confirmatory	R2	8	8	8	1	1	uniform_full

10 Discussion

Temporary cue-bearing full-support distributions changed whether or when the Transformer trained under the prospectively frozen protocol formed the modular predictive center. The target conditional, marginals, irreducible noise, total pairwise predictable information, architecture, optimizer, and objective were held fixed. Persistence after cue removal indicates that the early distribution altered the learning path rather than supplying a permanently required feature. Heterogeneity across rules and geometries rejects an explanation based on the scalar cue index alone.

10.1 Possible mechanism: symmetry breaking and optimization accessibility

Pure synergy removes singleton predictive signal from the population distribution. In the tested parameterization, this condition was associated with weak and poorly aligned early gradients. Cue-bearing distributions introduced lower-order statistical asymmetries that may have changed early gradient alignment and optimization accessibility. One possible consequence is an earlier path

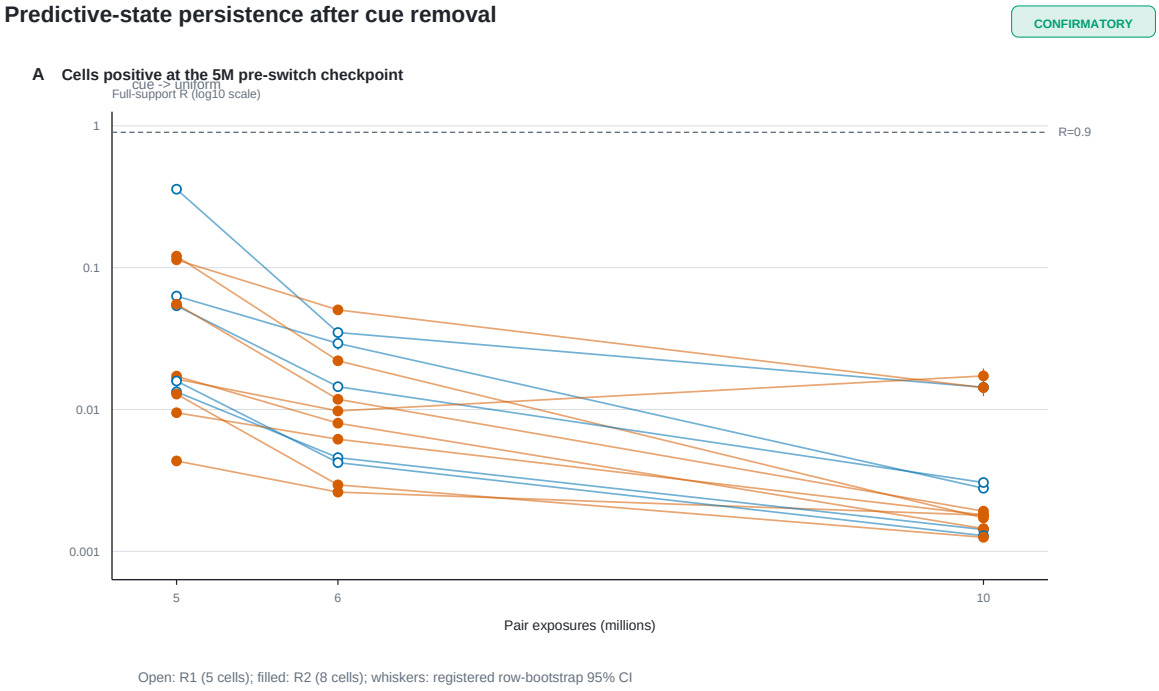


Figure 5: Persistence after return to uniform full support. Whiskers are registered first-parent-row bootstrap intervals.

toward composing both parents; after the task-sufficient representation forms, the cue is no longer required. This is an experiment-bounded interpretation, not a theorem that gradients generally vanish under pure synergy.

Three observations are consistent with this account: onset was ordered by registered cue dose, threshold-positive representations persisted after cue removal, and the frozen gradient audits showed changes in gradient coherence around observed transitions. This is mechanistic interpretation, not a causal mediation result. *Gradient statistics are correlates of transition dynamics, not identified causal mediators.* The data do not establish a universal loss-landscape mechanism or a unique circuit.

10.2 Practical implication: distributional bootstrapping

The result suggests a possible design principle: temporary training distributions can be used to improve access to otherwise hard-to-reach compositional representations when the final task distribution contains higher-order dependencies. This is a hypothesis for bounded, auditable interventions, not a claim that all AI training should use low-order cues.

11 Limitations

First, model scale is limited to one approximately 2M-parameter Transformer family; scale dependence remains open. Second, the parent-exposed interface intentionally controls variable identification; recurrent context discovery and variable selection remain future work. Third, exact soft-label

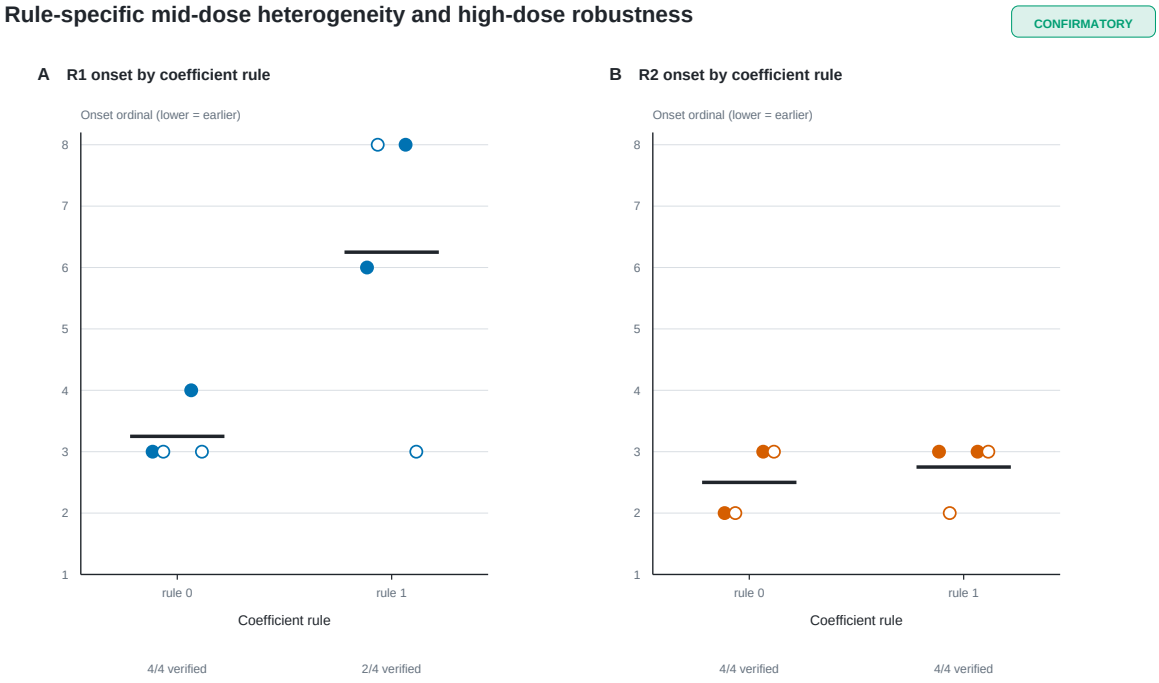


Figure 6: Rule-specific R1 heterogeneity and R2 robustness.

training isolates representation formation from sampled-label variance; sampled-token training remains future work. Fourth, finite-field modular arithmetic is a controlled model organism rather than a proxy for the full range of natural-data learning. Fifth, the experiments identify a distribution intervention but do not establish a universal loss-landscape mechanism.

The study also uses one field size, one noise level, a finite checkpoint grid, and eight confirmatory blocks. The exact test is valid for the frozen randomization structure but does not make the tested rules a population sample of all modular tasks. The 0–5M cue window coincided with the prospectively frozen 5M learning-rate warmup, and cue timing was not crossed with optimizer phase. Cue conditions preserved both parent marginals but changed their joint distribution, so the intervention is not a marginal-frequency manipulation alone. Dependence on model scale, field size, noise level, input interface, and exact soft labels remains open. The result is not a universal Transformer claim, not a thermodynamic phase transition, and not evidence that the recurrent production task is solved. These results are excluded from the original scaling fit.

12 Reproducibility statement

The confirmatory result and its frozen hashes are recorded in the supplementary artifact package. Frozen hashes, environment, test definitions, counts, and provenance are indexed in the package. The lightweight verifier recomputes distribution audits, transition coding, all 1,679,616 exact assignments, figures, and tables without optimizer steps.

Tables 7 and 8 summarize protocol integrity and claim boundaries.

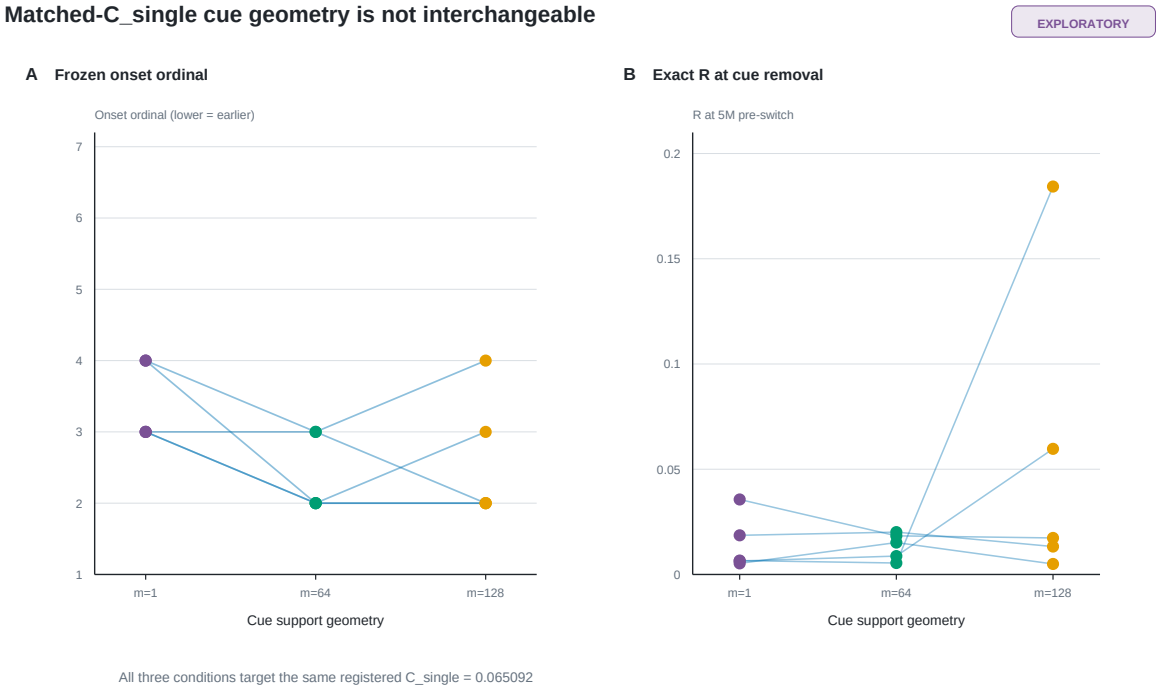


Figure 7: Exploratory matched-index geometry comparison.

Table 7: Counts and integrity flags are read from the frozen result and confirmatory freeze records.

Version	Status	Registered runs	Completed runs	Checkpoints	Sidecars valid	Evaluations	Probe reports	Gradient audits	Switch checks	Switches passed	Max exposure	Objective	Excluded from scaling fit
v3.5	exploratory	30	30	330	yes	360	210	150	30	yes	10000000	exact_soft_label_cross_entropy	yes
v3.6	confirmatory	24	24	264	yes	288	168	0	24	yes	10000000	exact_soft_label_cross_entropy	yes

13 Data and code availability

Code, manifests, result JSON, source data, and verification scripts are included in the repository. Checkpoints are referenced by SHA-256 sidecars and are unnecessary for lightweight verification.

14 Supplementary methods

14.1 Task objective, evaluation, and probes

Exact soft-label cross-entropy was used throughout the parent-exposed experiments: at each observed pair, the target was the analytic noisy conditional rather than one sampled label. Evaluation enumerated all 66,049 ordered parent pairs. Every checkpoint recorded exact conditional KL, normalized R, center accuracy, center probability, predictive entropy, and a 2,000-replicate

Table 8: The manuscript uses the allowed scoped statements and excludes the listed overclaims.

Topic	Allowed statement	Excluded statement
Causal scope	The registered transient cue-bearing full-support distributions changed onset timing or probability in the frozen parent-exposed task and model family.	Singleton mutual information alone is the causal variable.
Persistence	Threshold-positive representations persisted after return to uniform full support.	The cue is permanently required for prediction.
Generality	The ordered effect replicated across the two new rules and two new permutations tested.	This is a universal Transformer property.
Dynamics	The registered checkpoints show abrupt or delayed representation formation in some trajectories.	The result is a thermodynamic phase transition.
Recurrent task	The parent-exposed task isolates a mechanism relevant to the broader program.	The recurrent production task has been solved.
Scaling	The experiments are mechanism-focused exploratory and confirmatory audits.	The result belongs in the original scaling fit.
C_single	C_single is a registered scalar cue index used to order selected distributions.	C_single is a unique-information percentage.

first-parent-row bootstrap interval. Selected checkpoints included layer-wise linear, fixed-Fourier, and fixed-capacity MLP probes.

14.2 Cue-family construction and audit

Frozen cue permutations and coefficients reweighted the complete pair space while preserving strictly positive probability for every pair. Lambda was solved by deterministic root finding for each prospectively frozen target, not selected from model outcomes. Exhaustive enumeration verified marginal uniformity, conditional invariance, joint information, and full support.

14.3 Model and optimization

The v3.6 confirmatory manifest instantiated the architecture and optimization settings frozen from v3.5. It records 2,045,568 core parameters, width 168, batch size 4,096, exact soft-label cross-entropy, AdamW learning rate 3×10^{-4}, betas $(0.9, 0.95)$, weight decay 0.1, clipping at 1.0, 5M warmup exposures, and an absolute-exposure cosine schedule.

14.4 Transition coding

Probe checkpoints were 2M, 3M, 4M, 5M, 6M, 7.5M, and 10M. Onset required $R < 0.9$ and layer-6 linear accuracy above 0.5, retained at the next checkpoint. Establishment required $R < 0.1$, center accuracy above 0.9, and layer-6 linear accuracy above 0.9.

14.5 Exact test

Within-block terminal onset ordinals were converted to doubled midranks, with earlier onset ranked higher. All six label assignments were combined across eight blocks, giving $6^8 = 1,679,616$ assignments. Under the sharp null, the R0/R1/R2 labels were exchangeable within each block. The one-sided upper tail included the observed assignment.

14.6 Artifact integrity

The freeze records every tag, manifest and result hash, the environment lock, test definition and result, and artifact counts. This manuscript revision changes only the `paper/` tree; no prior scientific artifact changes.

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